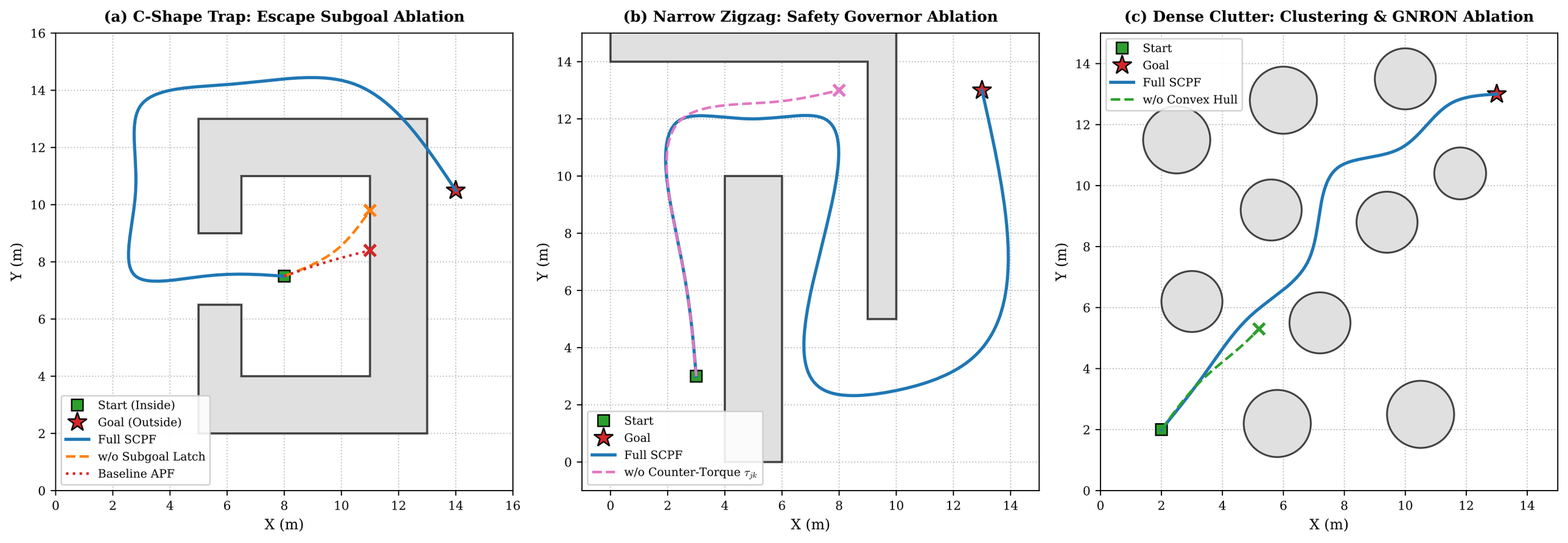
(a) C-Shape Trap: Escape Subgoal Ablation
Start (Inside)
Goal (Outside)
Full SCPF
w/o Subgoal Latch
Baseline APF
X (m)
Y (m)
(b) Narrow Zigzag: Safety Governor Ablation
Start
Goal
Full SCPF
w/o Counter-Torque τ_{jk}
X (m)
Y (m)
(c) Dense Clutter: Clustering & GNRON Ablation
Start
Goal
Full SCPF
w/o Convex Hull
X (m)
Y (m)